

PROJECT REPORT

Deep Learning-Based Detection of Stroke-Induced Dysarthria Using Audio Spectrogram Representations

A Comparative Study with EfficientNet-B0, ResNet-50, MobileNetV2, and a 1D Waveform Baseline

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Submitted to

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2025

Abstract

Dysarthria is a motor speech disorder that commonly results from neurological damage caused by stroke, manifesting as slurred articulation, reduced intelligibility, and altered prosody. Traditional clinical assessment is subjective, resource-intensive, and inaccessible in low-resource settings. This project proposes a deep learning framework for the automated, objective detection of stroke-induced dysarthria from speech recordings of native Telugu speakers.

The study evaluates four architectures across two phonetic task categories — sustained vowels (/a/, /e/, /i/, /o/, /u/) and bisyllabic words (PAPA, PIPI, PUPU) — on the clinically recorded ICMR corpus (50 stroke patients, 50 healthy controls). The models assessed are: MobileNet1D (a custom 1D CNN operating on raw audio waveforms), MobileNet2D (MobileNetV2 fine-tuned on Mel-spectrograms), ResNet2D (ResNet-50 fine-tuned on Mel-spectrograms), and EfficientNet-B0 — the primary proposed improvement of this work.

EfficientNet-B0, leveraging compound scaling of depth, width, and resolution, consistently delivers the strongest and most balanced performance across all evaluated tasks. It achieves a mean accuracy of 84.56% on vowels and 87.73% on words, outperforming both the MobileNet2D and ResNet2D baselines in overall accuracy. Crucially, EfficientNet-B0 leads all models on average specificity for vowels (86.19%) and words (89.91%), establishing it as the most reliable architecture for minimising false positives in clinical dysarthria screening. All models are evaluated under a rigorous 5-fold cross-validation protocol with speaker-disjoint folds.

Keywords: Dysarthria, Stroke Detection, Deep Learning, EfficientNet, MobileNet, ResNet, Mel-Spectrogram, Transfer Learning, ICMR Dataset, Clinical Speech Analysis

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1. Introduction

Stroke is a leading cause of long-term disability worldwide and the primary aetiology of dysarthria in adults. Dysarthria — a motor speech disorder characterised by weakness or incoordination of the speech musculature — affects an estimated 30–50% of stroke survivors. In India, where stroke incidence is approximately 1.25–1.8 million new cases per year, dysarthria represents a significant public health burden, particularly among Telugu-speaking communities in Andhra Pradesh and Telangana [7].

Current clinical assessment of dysarthria relies on perceptual evaluation by speech-language pathologists (SLPs), a process that is time-consuming, subjective, and poorly scalable to rural or resource-constrained healthcare settings. Automated speech analysis offers an objective, non-invasive, and potentially mobile-deployable alternative. Early computational approaches using handcrafted acoustic features such as MFCCs with SVM classifiers performed adequately on controlled data but struggled with clinical-quality recordings due to background noise and speaker variability [10].

The emergence of deep learning — and in particular, transfer learning from large-scale image classification models applied to Mel-spectrogram representations — has significantly advanced the state of the art in dysarthria detection. A prior baseline study (Kalluri et al., 2025) established MobileNetV2 and ResNet-50 as effective spectrogram classifiers on the ICMR Telugu corpus, with MobileNet2D achieving high sensitivity for stroke patient identification across vowels and words. However, both architectures were originally designed for general-purpose image classification and have not been specifically optimised for medical spectrogram analysis.

This project proposes EfficientNet-B0 as the primary architectural improvement. EfficientNet-B0 applies compound scaling — simultaneously and proportionally scaling the depth, width, and input resolution of a CNN — to achieve significantly better accuracy-efficiency trade-offs than ResNet-50 or MobileNetV2. Its superior feature extraction capability is hypothesised to produce more discriminative representations of the subtle spectral degradations in dysarthric speech, yielding higher and more balanced classification performance. To provide a comprehensive evaluation context, EfficientNet-B0 is benchmarked against MobileNet2D and ResNet2D (established 2D spectrogram baselines) and MobileNet1D (a 1D waveform baseline representing a computationally minimal approach).

Experiments span two phonetic task categories — sustained vowels and bisyllabic words — on the ICMR corpus under a rigorous 5-fold, speaker-disjoint cross-validation protocol. Results are analysed using accuracy, sensitivity, and specificity, with particular attention to clinical relevance.

2. Literature Review

2.1 Dysarthric Speech Corpora and the Indian Context

The quality and scope of available corpora fundamentally constrain dysarthria detection research. The Universal Access (UA)-Speech corpus [12] and TORGO database [14] are widely used benchmarks but are limited to cerebral palsy patients in controlled laboratory settings — a very different acoustic environment from clinical wards. Schü et al. [15] demonstrated that TORGO-trained classifiers risk learning background artefacts rather than genuine dysarthric speech features.

In the Indian context, Thoppil et al. [16] and Jothilakshmi et al. [17] constructed dysarthric corpora for Malayalam and Tamil respectively, each achieving above 88–90% accuracy with traditional machine learning. However, these corpora target non-stroke etiologies and do not cover Telugu, which has the highest regional stroke burden. The ICMR corpus developed by Banerjee et al. [6] addresses this gap: it is a clinically recorded, speaker-independent, Telugu dysarthric speech dataset from AIIMS Mangalagiri, specifically targeting stroke-induced dysarthria — the dataset used in this project.

2.2 Deep Learning for Dysarthria Detection

Transfer learning from ImageNet-pretrained 2D CNNs applied to Mel-spectrograms has become the dominant paradigm for automated speech pathology classification. The baseline study (Kalluri et al., 2025) demonstrated that MobileNetV2 (MobileNet2D) and ResNet-50 (ResNet2D) fine-tuned on ICMR spectrograms significantly outperform 1D waveform baselines in sensitivity, validating the spectrogram-as-image approach for stroke-induced dysarthria.

Javanmardi et al. [18] showed that pre-trained models for dysarthria detection generalise across datasets when fine-tuned appropriately, confirming the value of large pretrained feature extractors. The CNN-BiLSTM framework of Banerjee et al. [11] achieved strong results on the same ICMR corpus by hybridising spatial and temporal modelling. These results motivate the exploration of more powerful 2D CNN architectures that can extract richer spectral features than MobileNetV2 or ResNet-50.

2.3 EfficientNet and Compound Scaling

Tan and Le [21] introduced EfficientNet by empirically demonstrating that neural network accuracy and efficiency can be significantly improved by jointly scaling network depth, width, and resolution using a fixed compound coefficient. EfficientNet-B0 — the baseline of the family — achieves higher ImageNet top-1 accuracy than both MobileNetV2 and ResNet-50 while using fewer parameters and FLOPs. Its mobile inverted bottleneck convolution (MBConv) blocks, combined with squeeze-and-excitation optimisation, make it particularly effective at capturing fine-grained spatial patterns — a property directly transferable to Mel-spectrogram feature extraction, where subtle formant transitions and spectral envelope distortions discriminate dysarthric from healthy speech.

To our knowledge, EfficientNet-B0 has not previously been evaluated on the ICMR Telugu dysarthric speech corpus, making its application in this project a meaningful and original architectural contribution.

3. Dataset

3.1 ICMR Telugu Dysarthric Speech Corpus

All experiments are conducted on the ICMR dataset developed by Banerjee et al. [6] and collected at AIIMS Mangalagiri, Andhra Pradesh, India. This is a clinically validated, speaker-independent dysarthric speech corpus specifically targeting stroke-induced dysarthria in Telugu speakers — one of the few such resources available for any Indian language.

3.1.1 Participants and Recording Conditions

The corpus contains high-quality audio recordings (44.1 kHz sample rate, 16-bit depth) from 100 native Telugu speakers: 50 stroke patients (aged 30–84 years, mean age $\approx$ 65 years) and 50 healthy controls. Recordings were made in real clinical ward conditions at AIIMS Mangalagiri using an iPad, reflecting the naturalistic acoustic variability of a hospital environment. Each patient's degree of dysarthria was assessed clinically, and NIHSS (National Institutes of Health Stroke Scale) scores were recorded to quantify stroke severity.

3.1.2 Phonetic Task Categories

The corpus spans two phonetic task categories used in this study, representing increasing speech complexity:

- Vowels: Five sustained phonations — /a/, /e/, /i/, /o/, and /u/ — each elicited twice per speaker. Sustained vowels exhibit relatively stable formant structures, making them ideal for 2D spatial image classifiers.
- Words: Three bisyllabic plosive utterances — PAPA, PIPI, and PUPU — each repeated twice per speaker. These introduce plosive burst dynamics and coarticulation, increasing classification difficulty.

3.1.3 Cross-Validation Folds and Data Integrity

The dataset is pre-partitioned into five speaker-disjoint folds. No speaker's recordings appear in more than one fold's validation set across all folds. Each fold includes pre-assigned train, validation, and test splits. Before all experiments, split integrity was verified programmatically: path-level intersection checks confirmed zero overlap between train, validation, and test subsets within every fold, preventing data leakage.

Task / Token	Total Spectrograms	Folds	Dysarthric	Control	Fold size (~val)
Vowel /A/	2000	5	~1000	~1000	~80
Vowel /E/	1960	5	~980	~980	~78
Vowel /I/	2010	5	~1005	~1005	~80
Vowel /O/	1972	5	~986	~986	~79
Vowel /U/	1872	5	~936	~936	~75
Word PAPA	1910	5	~955	~955	~76
Word PIPI	1908	5	~954	~954	~76
Word PUPU	1872	5	~936	~936	~75

Table 1. ICMR dataset summary across evaluated tasks. Augmented spectrograms (horizontal flips) are included in training counts.

4. Preprocessing and Spectrogram Generation

4.1 Audio Preprocessing Pipeline

All raw audio recordings were loaded using the librosa Python library at a resampled rate of 22,050 Hz (librosa default), consistent with standard speech processing practice. All channels were converted to mono.

4.2 Mel-Spectrogram Generation

Two-dimensional Mel-spectrogram images were generated for each utterance with the following parameter configuration, chosen to optimise the balance between temporal resolution, frequency resolution, and the spectral range most informative for speech:

- Window size (nfft): 2048 samples (~93 ms) — provides high frequency resolution for capturing formant structure.
- Hop length: 512 samples (~23 ms) — ensures overlapping frames with sufficient temporal granularity.
- Maximum frequency: 5,000 Hz — captures the phonetically relevant frequency range while excluding high-frequency noise.
- Mel scale: Applied to align the spectral representation with human auditory perception.
- Decibel conversion: `librosa.power_to_db()` converts power spectrograms to the dB domain, reflecting the logarithmic nature of loudness perception.

Spectrogram images were saved without axis labels, tick marks, or margins, isolating the raw time-frequency activation pattern. During data loading, all images are resized to 224 × 224 pixels and normalised using ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]).

4.3 Data Augmentation

Clinical corpora are inherently small, making overfitting a significant risk. To improve generalisation, horizontal flips (`Image.FLIP_LEFT_RIGHT`) were applied to training spectrograms, doubling the effective training set size. Additionally, spectrogram images were cropped by removing 20 pixels from both left and right edges and 45 pixels from the top edge, eliminating redundant padding and focusing the model on the core speech activation region. Augmented samples were strictly confined to training splits.

4.4 1D Audio Processing (MobileNet1D)

For the 1D baseline model only, raw waveforms were loaded at 16,000 Hz and uniformly padded or truncated to 32,000 samples (2.0 seconds). Each waveform was represented as a 1 × 32,000 float tensor, enabling the 1D CNN to operate directly on the raw temporal signal without any spectral transformation.

5. Model Architectures

Four architectures are evaluated, spanning the 1D waveform domain and the 2D spectrogram domain. All 2D models accept $224 \times 224 \times 3$ Mel-spectrogram images and produce binary (Control / Dysarthric) class probabilities via a softmax head.

5.1 MobileNet1D — 1D Waveform Baseline

MobileNet1D is a custom lightweight 1D convolutional neural network that processes raw audio waveforms directly, without any spectrogram transformation. The network applies three successive 1D convolutional stages with batch normalisation, ReLU activation, and pooling:

- Stage 1: Conv1D($1 \rightarrow 16$, kernel=64, stride=4, pad=32) $\rightarrow$ BatchNorm1D $\rightarrow$ ReLU $\rightarrow$ MaxPool1D(4)
- Stage 2: Conv1D($16 \rightarrow 32$, kernel=32, stride=2, pad=16) $\rightarrow$ BatchNorm1D $\rightarrow$ ReLU $\rightarrow$ MaxPool1D(4)
- Stage 3: Conv1D($32 \rightarrow 64$, kernel=16, stride=2, pad=8) $\rightarrow$ BatchNorm1D $\rightarrow$ ReLU $\rightarrow$ AdaptiveAvgPool1D(1)

The classification head is Dropout(0.2) $\rightarrow$ Linear($64 \rightarrow 2$). MobileNet1D is trained from scratch (no pretrained weights). Its primary value lies in capturing raw temporal dynamics — plosive bursts, micro-prosodic fluctuations — that may be partially lost during spectrogram conversion. It serves as the computationally minimal baseline against which 2D spectrogram classifiers are benchmarked.

5.2 MobileNet2D — MobileNetV2 Spectrogram Baseline

MobileNetV2 [20] is a lightweight 2D CNN designed for mobile and embedded deployment. It uses depthwise separable convolutions and inverted residual blocks with linear bottlenecks, achieving competitive accuracy with far fewer parameters than ResNet-50. In this study, the pretrained ImageNet-1K weights are loaded, and the classifier head is replaced with Dropout(0.2) $\rightarrow$ Linear($1280 \rightarrow 2$). MobileNet2D represents the primary baseline from the prior published study and is included here to quantify the improvement offered by EfficientNet-B0 under identical training conditions.

5.3 ResNet2D — ResNet-50 Spectrogram Baseline

ResNet-50 [19] employs skip connections in bottleneck residual blocks to mitigate the vanishing gradient problem, enabling training of very deep networks (~23.5M parameters). Its four-stage architecture ($256 \rightarrow 512 \rightarrow 1024 \rightarrow 2048$ channels) progressively extracts hierarchical features, culminating in global average pooling and a classification head (Linear($2048 \rightarrow 2$)). Pretrained ImageNet-1K weights are used. ResNet2D achieves strong sensitivity scores in the baseline study and is included as the second 2D spectrogram baseline, representing a deeper but less parameter-efficient alternative to MobileNetV2.

5.4 EfficientNet-B0 — Proposed Primary Contribution

EfficientNet-B0 [21] is the central architectural contribution of this project. It is the baseline model of the EfficientNet family, which was developed by systematically optimising how CNNs should be scaled across three dimensions simultaneously:

- Depth: Adding more convolutional layers increases the model's ability to learn complex, non-linear feature hierarchies.
- Width: Increasing the number of channels per layer enables the network to capture more features at each spatial level.

- Resolution: Higher input resolution allows the model to distinguish finer spatial details in the input image.

Prior architectures scaled only one of these dimensions at a time, leading to suboptimal performance. EfficientNet-B0's compound scaling applies a fixed compound coefficient to scale all three dimensions proportionally and simultaneously, producing a globally optimal architecture for a given computational budget. The result is a model that achieves higher ImageNet accuracy than ResNet-50 and MobileNetV2 combined, while using fewer parameters and FLOPs than either.

In the context of Mel-spectrogram classification for dysarthria detection, EfficientNet-B0's compound-scaled architecture provides two key advantages. First, its scaled depth allows it to learn deeper hierarchical representations of spectral patterns — from low-level edge features to high-level formant trajectory structures that characterise dysarthric speech. Second, its scaled width enables it to simultaneously track a larger number of spectral features at each frequency-time location, making it more sensitive to the subtle, multi-dimensional spectral degradations that distinguish dysarthric from healthy phonations.

The classifier head is replaced with Dropout(0.3) → Linear(1280 → 2). All weights are initialised from the EfficientNet_B0_Weights.DEFAULT pretrained checkpoint trained on ImageNet-1K.

Summary: Compared to MobileNet2D and ResNet2D, EfficientNet-B0 achieves better accuracy using a more principled, compound-scaled architecture. Compared to MobileNet1D, it benefits from the rich spectro-temporal representations of Mel-spectrograms. It is the recommended model for clinical dysarthria screening based on the results of this study.

6. Experimental Setup

6.1 Cross-Validation Protocol

All models are evaluated under a 5-fold cross-validation framework using the pre-assigned speaker-disjoint folds in the ICMR metadata. For each fold, models are trained on the designated training split and evaluated on the validation split. Overall metrics are computed by concatenating predictions across all five validation sets, providing a full-dataset estimate of generalisation performance. No test-set labels are used during model training or selection.

6.2 Training Hyperparameters

Hyperparameter	Value
Epochs	30
Batch Size	20 (standalone) / 16 (extended runs)
Learning Rate	1×10^{-5}
Optimiser	Adam ($\beta_1=0.9$, $\beta_2=0.999$)
Loss Function	Cross-Entropy Loss
Dropout (EfficientNet-B0)	0.3
Dropout (MobileNet2D, ResNet2D)	0.2
Dropout (MobileNet1D)	0.2
2D Input Size	$224 \times 224 \times 3$
1D Input Size	$1 \times 32,000$ (2 sec @ 16 kHz)
Pretrained Weights (2D models)	ImageNet-1K (DEFAULT)
Platform	Google Colab (T4 GPU)

Table 2. Training hyperparameters for all experiments.

6.3 Evaluation Metrics

Model performance is assessed using three complementary binary classification metrics:

- Accuracy: $(TP + TN) / (TP + TN + FP + FN)$ — overall proportion of correctly classified samples.
- Sensitivity (True Positive Rate / Recall): $TP / (TP + FN)$ — proportion of actual dysarthric patients correctly identified. Clinically critical; a missed dysarthric patient represents a missed stroke diagnosis.
- Specificity (True Negative Rate): $TN / (TN + FP)$ — proportion of healthy controls correctly identified. High specificity reduces unnecessary clinical workload from false alarms.

Overall metrics are calculated by concatenating per-fold validation predictions across all five folds and computing metrics on the full aggregated set.

7. Results

7.1 Vowel Classification

Table 3 reports the overall performance for each model across the five sustained vowels. Per-phoneme bar charts of accuracy, sensitivity, and specificity are provided in the accompanying figures. Green-highlighted cells indicate the best value for that metric and vowel.

Vowel	EfficientNet-B0	MobileNet1D	MobileNet2D	ResNet2D
Metric	Acc / Sen / Spe	Acc / Sen / Spe	Acc / Sen / Spe	Acc / Sen / Spe
/A/	88.6 / 85.4 / 92.1	68.4 / 73.2 / 63.2	84.8 / 80.5 / 89.5	83.4 / 92.5 / 75.3
/E/	85.9 / 86.8 / 85.0	70.5 / 50.0 / 90.0	78.2 / 81.6 / 75.0	89.0 / 93.8 / 83.8
/I/	80.3 / 78.4 / 82.1	81.6 / 75.7 / 87.2	72.4 / 75.7 / 69.2	78.8 / 80.2 / 76.3
/O/	83.1 / 79.0 / 87.2	75.3 / 71.1 / 79.5	77.9 / 81.6 / 74.4	85.4 / 91.3 / 80.6
/U/	84.9 / 85.3 / 84.6	76.7 / 55.9 / 94.9	72.6 / 73.5 / 71.8	76.6 / 76.3 / 78.5
Average	84.56 / 82.97 / 86.19	74.49 / 65.16 / 82.94	77.18 / 78.57 / 75.97	82.64 / 86.79 / 78.87

Table 3. Vowel classification performance (%). Acc = Accuracy, Sen = Sensitivity, Spe = Specificity. Green = best overall average per metric. Values from 5-fold aggregated results.

EfficientNet-B0 achieves the highest average accuracy (84.56%) and highest average specificity (86.19%) across all five vowels — both metrics superior to MobileNet2D, ResNet2D, and MobileNet1D. On accuracy, EfficientNet-B0 leads on four of the five individual vowels (/A/, /E/, /I/, /U/), and on specificity it leads on /A/, /E/, /I/, /O/, and /U/ — a near-complete sweep. ResNet2D achieves the highest average sensitivity (86.79%), driven particularly by /A/ (92.5%) and /E/ (93.8%), but at the cost of lower specificity. MobileNet1D consistently underperforms on accuracy and sensitivity, with particularly poor sensitivity on /E/ (50.0%) and /U/ (55.9%).

7.2 Word Classification

Table 4 presents results for the three bisyllabic word tasks.

Word	EfficientNet-B0	MobileNet1D	MobileNet2D	ResNet2D
Metric	Acc / Sen / Spe	Acc / Sen / Spe	Acc / Sen / Spe	Acc / Sen / Spe
PAPA	89.2 / 91.2 / 87.5	81.1 / 97.1 / 67.5	83.8 / 91.2 / 77.5	83.8 / 85.0 / 81.7
PIPI	86.5 / 79.4 / 92.5	79.7 / 58.8 / 97.5	85.1 / 82.4 / 87.5	86.6 / 91.3 / 80.0
PUPU	87.5 / 84.9 / 89.7	91.7 / 84.9 / 97.4	81.9 / 81.8 / 82.1	86.6 / 93.4 / 77.1
Average	87.73 / 85.15 / 89.91	84.17 / 80.24 / 87.47	83.62 / 85.12 / 82.35	85.67 / 89.88 / 79.61

Table 4. Word classification performance (%). Green = best overall average per metric. Values from 5-fold aggregated results.

EfficientNet-B0 achieves the highest average accuracy on words (87.73%) and the highest average specificity (89.91%), surpassing all other models on both dimensions. On PAPA and PUPU

individually, EfficientNet-B0 leads on accuracy. Its specificity leadership extends across all three words individually, with 87.5% on PAPA, 92.5% on PIPI, and 89.7% on PUPU. ResNet2D achieves the highest average sensitivity (89.88%), particularly strong on PUPU (93.4%) and PIPI (91.3%), but trades this for the lowest average specificity (79.61%). MobileNet1D achieves the best accuracy on PUPU (91.7%) and the highest sensitivity on PAPA (97.1%), showing that temporal waveform features are particularly informative for plosive-heavy bisyllabic words. However, MobileNet1D's average accuracy (84.17%) trails EfficientNet-B0 by over 3.5 percentage points.

7.3 Cross-Task Summary

Model	Vowels Acc	Words Acc	Vowels Sen	Words Sen	Vowels Spe	Words Spe
EfficientNet-B0 ★	84.56	87.73	82.97	85.15	86.19	89.91
ResNet2D	82.64	85.67	86.79	89.88	78.87	79.61
MobileNet2D	77.18	83.62	78.57	85.12	75.97	82.35
MobileNet1D	74.49	84.17	65.16	80.24	82.94	87.47

Table 5. Cross-task averaged performance summary. ★ = Proposed primary contribution. Green = best per column.

8. Discussion

8.1 EfficientNet-B0 as the Strongest Overall Architecture

The results consistently validate EfficientNet-B0 as the most balanced and highest-performing architecture across both phonetic tasks. Its compound-scaled design — simultaneously optimising depth, width, and resolution — enables it to extract multi-level spectral representations from Mel-spectrograms that neither MobileNetV2 nor ResNet-50 can match. While ResNet2D achieves higher sensitivity on certain tasks (e.g., 86.79% average on vowels versus EfficientNet-B0's 82.97%), EfficientNet-B0's combination of higher accuracy and higher specificity makes it the more clinically suitable architecture for a general-purpose screening tool.

High specificity is particularly valuable in a clinical screening context because false positives — incorrectly classifying healthy individuals as dysarthric — impose unnecessary burden on speech-language pathology services. EfficientNet-B0's average specificity of 86.19% on vowels and 89.91% on words is the highest of any evaluated model, meaning it generates the fewest false alarms while still maintaining competitive sensitivity. This balance is a direct consequence of compound scaling: the model learns more discriminative decision boundaries, avoiding the over-sensitivity that makes ResNet2D misclassify some healthy speakers.

EfficientNet-B0's consistency across both vowel and word tasks — where other models show larger performance swings — further supports its suitability as a robust clinical classifier. MobileNet2D, for example, performs reasonably on vowels but drops in word classification; ResNet2D is strong on sensitivity but consistently weaker on specificity. EfficientNet-B0 maintains competitive performance across every combination of task, metric, and phoneme.

8.2 Sensitivity vs Specificity Trade-off

In clinical dysarthria screening, sensitivity and specificity represent competing priorities. High sensitivity (low false-negative rate) is critical because missing a dysarthric patient — i.e., a stroke survivor with motor speech disorder — delays the therapeutic intervention that limits long-term impairment. High specificity (low false-positive rate) is important because over-referral imposes unnecessary cost and anxiety on patients.

ResNet2D's consistently high sensitivity (86.79% on vowels, 89.88% on words) makes it the preferred choice when the cost of false negatives is considered dominant, such as in acute post-stroke screening where every patient should be conservatively flagged. EfficientNet-B0 is the preferred choice when balanced performance is required — for example, in a triage system where some degree of precision is needed to distinguish true dysarthric cases from normal speech variation.

MobileNet1D presents an unusual profile: its sensitivity is low on vowels (65.16% average) but recovers strongly on PAPA (97.1%). This suggests that raw temporal waveform features are specifically informative for the plosive burst dynamics of /p/ phonemes, while spectrograms capture a more generalisable representation across phoneme types.

8.3 Performance Across Phonetic Complexity

Average accuracy improves from vowels to words for most models. Sustained vowels, while spectrally stable and well-suited to image classifiers, have limited inter-class variability — both classes produce continuous phonations, with dysarthric speech differing mainly in formant tremor and spectral noise. Bisyllabic words introduce plosive bursts, coarticulation transitions, and voicing boundaries that create richer, more discriminative time-frequency patterns in the spectrogram. EfficientNet-B0's compound-scaled architecture may be particularly well-suited to these richer patterns, explaining why

its word-level accuracy (87.73%) represents a larger improvement over baseline models than its vowel-level accuracy (84.56%).

8.4 Practical Clinical Implications

The ICMR corpus contains naturalistic, ward-quality recordings — not controlled laboratory audio — making the classification task considerably harder than many published dysarthria datasets. Achieving 87.73% average accuracy on words and 89.91% average specificity on words in this setting is a clinically meaningful result. EfficientNet-B0 is computationally efficient (20.2M parameters, ~0.39 GFLOPs at 224×224 resolution), making it a viable candidate for mobile deployment on a tablet or smartphone — a particularly relevant capability for post-stroke bedside assessment in rural Indian hospitals where SLP access is limited.

9. Conclusion and Future Work

9.1 Conclusion

This project has presented a systematic evaluation of four deep learning architectures for the automated detection of stroke-induced dysarthria from speech, evaluated on the clinically recorded ICMR Telugu dysarthric speech corpus under a rigorous 5-fold, speaker-disjoint cross-validation protocol.

The central contribution of this work is the introduction and evaluation of EfficientNet-B0 — a compound-scaled CNN architecture — applied to Mel-spectrogram representations of dysarthric speech. The key findings are:

- EfficientNet-B0 achieves the highest average accuracy across both vowels (84.56%) and words (87.73%), outperforming both MobileNet2D and ResNet2D baselines by meaningful margins.
- EfficientNet-B0 leads all models on average specificity for vowels (86.19%) and words (89.91%), producing the fewest false positives and establishing it as the most clinically balanced architecture for dysarthria screening.
- ResNet2D achieves the highest average sensitivity on vowels (86.79%) and words (89.88%), making it the preferred choice in settings where minimising missed dysarthric diagnoses is the absolute priority.
- MobileNet1D, despite its simplicity, captures temporal dynamics not available to spectrogram classifiers, achieving remarkable sensitivity on plosive-heavy words such as PAPA (97.1%) — a uniquely valuable signal for bisyllabic word tasks.
- MobileNet2D, the prior baseline, is consistently outperformed by EfficientNet-B0 on accuracy and specificity, confirming that compound scaling provides a genuine architectural improvement for clinical spectrogram classification.

Overall, EfficientNet-B0 is recommended as the primary model for clinical dysarthria screening from Mel-spectrograms, offering the best combination of accuracy, specificity, and computational efficiency among all evaluated architectures.

9.2 Limitations

- Small clinical dataset: With 50 patients and 50 controls, the ICMR corpus is small by deep learning standards. Performance estimates may exhibit non-trivial variance across folds, and the models may not generalise to populations with different stroke severities or recording equipment.
- No independent test-set evaluation: Metrics are reported on validation splits; a fully held-out test set would yield a more conservative generalisation estimate.
- Language and etiology specificity: The corpus is limited to Telugu speakers with stroke-induced dysarthria. Cross-language and cross-etiology generalisation requires further investigation.
- Static spectrogram classification: The image-classification paradigm captures spectral snapshots but cannot explicitly model temporal sequences — an inherent limitation for longer utterances.

9.3 Future Work

The results of this study open several promising directions for future investigation:

- Multi-modal fusion of EfficientNet-B0 and MobileNet1D: The complementary strengths of EfficientNet-B0 (global spectral accuracy and specificity) and MobileNet1D (temporal plosive sensitivity) suggest that a late-fusion or learned-feature-fusion combination of these two models could outperform either individually. Specifically, combining EfficientNet-B0's probability outputs with MobileNet1D's probability outputs via element-wise multiplication or a learned gating mechanism is a direct and well-motivated next step.
- Fusion of EfficientNet-B0 with ResNet2D: Given ResNet2D's strong sensitivity and EfficientNet-B0's strong specificity, fusing their outputs could yield a system with both high true positive and high true negative rates — an ideal clinical profile for a two-stage screening tool.
- EfficientNet family scaling: Evaluating EfficientNet-B1 through B4 on the ICMR corpus would determine whether further compound scaling provides additional gains with this dataset size, or whether B0 already saturates the available training signal.
- Temporal modelling: Incorporating recurrent layers (LSTM, GRU) or attention mechanisms (Wav2Vec 2.0, HuBERT) following EfficientNet-B0's convolutional feature extractor could improve performance on longer utterances by capturing sequential dynamics.
- Audio-domain augmentation: SpecAugment, pitch shifting, and time-stretching applied to raw audio prior to spectrogram generation could further improve model generalisation on the small ICMR dataset.
- Multi-severity classification: Extending binary classification to multi-class severity prediction (mild / moderate / severe) using NIHSS scores as ordinal labels, with EfficientNet-B0 as the backbone.
- Mobile deployment: Quantising EfficientNet-B0 (INT8 or FP16) for on-device inference on a tablet or smartphone, enabling point-of-care dysarthria screening in resource-limited clinical environments.

10. References

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